

Channel-wise Contribution Assessment for RGB-D Salient Object Detection

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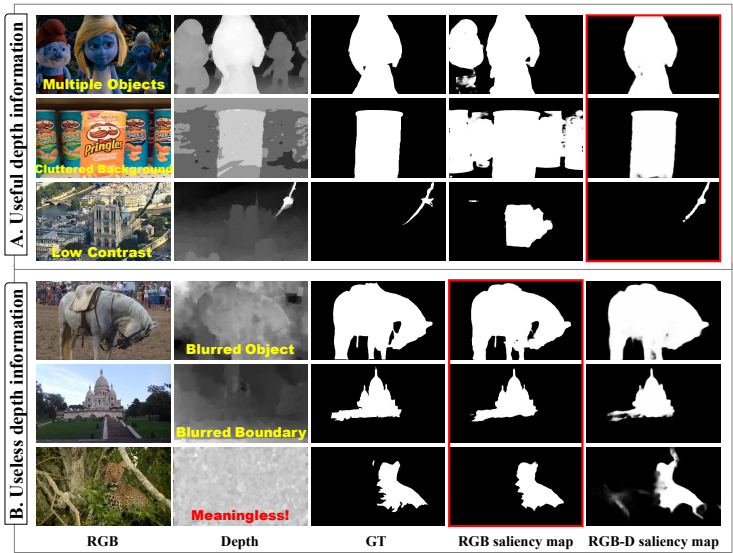


Fig. 1. In complex scenes (A), RGB channels may contribute less to accurate salient object detection compared to depth information, which contains valuable and informative data. Solely relying on RGB may not yield precise saliency results. Conversely, when depth channels exhibit a lower contribution (B), it can adversely affect the corresponding RGB channels in RGB-D SOD. The red rectangular illustrates how emphasizing the high-contribution modality can result in superior saliency outcomes in these scenarios.

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In RGB-D salient object detection (SOD), a common approach to improving accuracy is by using a dual-stream architecture to combine RGB and depth data. However, the effectiveness of depth information varies depending on the scene. In scenarios where depth maps provide a limited contribution, integrating them with RGB can be challenging and sometimes even detrimental to performance rather than enhancing it. Conventional RGB-D SOD methods often lack precision in assessing depth map quality, neglecting to account for the distinct contributions of various regions within the map, often resulting in a suboptimal fusion of regions where depth information is minimally beneficial or irrelevant. To address these issues, this paper presents a novel channel-wise contribution assessment method that precisely evaluates the contributions of both the RGB and depth channels. By employing a controlled perturbation process to challenge the saliency detection model with specific, manageable disturbances, we are able to measure how much RGB and depth information each contributes to the final saliency map. Based on this analysis, we have developed a novel routing-style fusion of modality that dynamically adjusts the integration of the two modalities. This approach significantly lessens the negative impact of regions where depth data have a low, no, or even detrimental contribution, leading to a more effective and balanced fusion of RGB and depth information. Extensive experiments on multiple benchmark datasets demonstrate that the proposed method consistently achieves competitive performance and improves the robustness of RGB-D salient object detection across diverse and challenging scenarios.

CCS Concepts: • **Computing methodologies** → **Object detection**.

Additional Key Words and Phrases: RGB-D Salient Object Detection, Channel Contribution-aware, Modality Routing Fusion.

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1 Introduction

Salient object detection (SOD) [39, 40] simulates human visual attention mechanism, aiming to quickly locate and segment the most significant objects in an image. This task is crucial in the field of computer vision and plays a vital role in various applications, including image retrieval [10], image reconstruction [35], video analysis [39], quality assessment [6], and visual tracking [26].

Leveraging the strong capabilities of deep learning in capturing intricate patterns, the field of salient object detection has seen remarkable progress recently, as highlighted by studies such as [29, 52, 65]. Despite these advances, detecting salient objects in RGB images alone faces challenges in complex scenarios like scenes with multiple objects, cluttered backgrounds, and images with low contrast, as illustrated in Fig. 1-A. These challenges underscore the necessity for incorporating depth maps (D) that supplement RGB images with valuable spatial details. In response, RGB-D salient object detection techniques, exemplified by [25, 42, 54, 57, 69, 70], employ a dual-stream strategy. This approach involves extracting color, texture, and semantic data from RGB images and spatial cues from depth maps, then merging these elements to pinpoint salient features within RGB-D contexts. Thus, attaining an optimal complementary status between RGB and depth is the primary challenge in RGB-D salient object detection. Although existing fusion-based methods [28, 38, 41, 59, 68] are adept at integrating RGB and depth information, they tend to fail when facing low-contribution D. As depicted in Fig. 1-B, the quality of depth data can significantly vary due to differences in depth sensors, often resulting in noisy or blurred depth maps. These low-quality depth maps contribute less to the learning process of RGB-D models, thus greatly degrading models' performance.

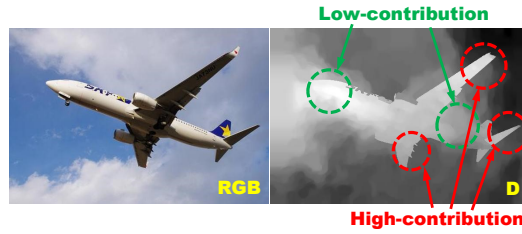


Fig. 2. Demonstration of regions within the depth channel that exhibit varying contributions. Detailed contribution analyses of the RGB channel and depth channel, especially the different regions within the depth channel, offer a precise perception of how these modalities complement each other.

Several researchers [4, 16, 51, 71, 72], have enhanced the performance of RGB-D salient object detection by focusing on depth maps' quality evaluation and optimization. For example, Fan et al. [16] developed a depth enhancement module to refine depth information, aiming to better merge RGB and depth features. Similarly, Wang et al. [51] created multi-level depth features to assess depth quality by identifying attributes of high-quality depth areas. Chen et al. [4] proposed a solution with a depth quality assessment subnet, selecting crucial depth areas for network training through pseudo-labels. While these depth quality evaluation/optimization method, to a certain extent, solve the problem of performance degradation caused by poor depth map quality, their effectiveness is limited in complex scenes. The reasons can be attributed to three aspects: 1) They assess or optimize the quality of the depth map in a simple or adaptive way, which may not fully capture important features of depth information in complex scenes, leading to the loss of crucial information and affecting the accuracy of the final detection results; 2) If the evaluation or optimization methods lack robustness, it may result in the model exhibiting instability when handling different types of depth maps, thereby reducing its generalization ability in practical scenarios; 3) They do not adequately consider the varying contributions of different channels¹, such as the RGB or depth channel. This is because different regions in depth maps contribute differently to saliency detection results, potentially introducing perceptual biases and the risk of generating misleading information.

In general, salient regions in images often have a defining feature: the high-contribution regions easily distinguish salient objects from the background, whereas low-contribution regions merge with, sharing similar colors and textures with their surroundings. As illustrated in Fig. 2, certain regions of a salient object can be clearly identified from the non-salient background using only the depth channel, as indicated by red arrows. Conversely, other areas are harder to differentiate, with blurred boundaries highlighted by green arrows. Interestingly, high-contribution regions maintain their saliency even when disrupted by noise, whereas low-contribution regions are more likely to “collapse”—the previously detected regions could be changed significantly (we will detail it in Fig. 4 latter). This observation leads us to propose a novel channel-wise contribution assessment approach to solve the above-mentioned three issues. The key idea of our method is to precisely evaluate the separate contributions of different modality channels, like RGB and D, within various “fine-grained” scene regions. Then, guided by the channel-wise contribution, we have presented a new routing-style fusion scheme to optimize the synergy between RGB and D channels. Unlike previous techniques, our approach uniquely balances the complementary status between RGB and D for effective fusion, bypassing the need for additional labeling or learning. This ensures that even in complex scenes, our method achieves a coherent and accurate salient object detection. Code and datasets are publicly available at <https://github.com/zhangmoon1/CCA-SOD>.

¹The term “channel” in this paper refers to the three-channel color image, i.e., R, G, B, or the single-channel depth map, i.e., D.

Overall, the main contributions of this paper can be summarized as follows:

- We present a new perspective for RGB-D salient object detection by explicitly modeling modality channel contributions. Rather than treating depth quality as a global property, our method emphasizes that the usefulness of RGB and depth information varies across scenes and regions, and should therefore be estimated in a fine-grained manner before fusion.
- We propose a saliency perturbation-based channel contribution assessment framework to estimate the contributions of RGB and depth channels without additional labeling or retraining. By combining channel-wise early fusion with controlled perturbations, the proposed method supports both coarse-grained and fine-grained contribution estimation, enabling more precise perception of modality complementarity.
- We develop a contribution-aware fusion scheme that uses the estimated channel contributions to guide the selection between RGB saliency and RGB-D saliency. This design effectively suppresses the interference of low-contribution depth cues and improves the complementary use of RGB and depth information in complex scenes.

2 Related Work

Existing fusion-based RGB-D salient object detection (RGB-D SOD) methods [5, 17, 19, 30], typically employ a dual-stream architecture that includes depth maps as an additional layer of input. These depth cues enrich the model with contextual and spatial information, enhancing the RGB stream's understanding of object locations and their interrelations. However, the effectiveness of the depth channel in complementing the RGB data is inconsistent, largely due to variations in depth map quality. To mitigate the influence of depth quality on detection performance, approaches generally fall into two categories: depth quality optimization and depth quality assessment. The former focuses on enhancing the depth maps themselves, while the latter involves evaluating and quantitatively measuring depth image quality to minimize the negative effects of poor-quality depth inputs. We'll look at these two types of methods in detail next.

2.1 Depth Quality Optimization in RGB-D SOD

Depth quality optimization techniques aim to enhance the clarity and usefulness of depth maps either through self-improvement or with the aid of RGB data, mitigating the effects of low-quality depth inputs. Self-enhancement strategies often leverage attention mechanisms to refine depth features. For instance, Fan et al. [16] developed a depth enhancement module that employs channel and spatial attention mechanisms to improve depth information before its integration with RGB data. On the other hand, RGB-assisted methods for enhancing depth information are more prevalent. A common strategy involves using RGB features to filter noise from depth data. Zhao et al. [63] introduced a network that uses RGB-based contrast loss to enhance depth feature quality. Chen et al. [8] generated rough saliency maps from RGB images to refine depth maps by emphasizing salient features and reducing background noise. Liao et al. [28] proposed using RGB features to identify and amplify significant elements in depth data through a cross-modal attention module.

HiDANet [53] uses a granularity-based attention scheme to enhance RGB and depth feature discrimination. DCF [21] designs a calibration module to improve the depth quality. Lee et al. [24] calculate the reliability of each modality at each stage, then combine them through their reliabilities. More recently, several studies have explicitly focused on handling low-quality depth maps in RGB-D SOD. For example, DCMNet [47] introduces a discriminative and cross-modality learning strategy to model depth reliability, while AMINet [50] employs attention-guided multi-modality interaction to alleviate the interference caused by unreliable depth cues. In addition, depth quality-inspired

feature manipulation has also been explored to adaptively enhance RGB-D feature integration and improve efficient salient object detection [62].

Despite these advancements, such methods do not fully account for the distinct contributions of each channel. This oversight could impact the fusion of RGB and depth information and potentially introduce perceptual biases and the risk of generating misleading information, underlining the need for approaches that recognize and balance the unique contribution of both data types.

2.2 Depth Quality Assessment in RGB-D SOD

Depth quality assessment techniques [3, 4, 9, 11, 15, 51] aim to measure the quality of depth maps or features, limiting the influence of poor-quality depth data. Early methods relied on handcrafted metrics developed from domain knowledge and heuristic evaluations. For instance, Wang et al. [51] crafted features reflecting the traits of high-quality depth regions and assessed these using a multi-scale method. Nonetheless, the manual approaches are often labor-intensive and struggle with complex scenarios due to their limited scope for characterizing depth quality. With the advent of deep learning, researchers have shifted towards using networks to assess depth quality, employing weakly supervised learning techniques. Chen et al. [4] introduced a weakly supervised subnet that identifies valuable depth regions as pseudo-labels for training by analyzing the congruence between saliency maps and the ground truth of RGB and depth data. Bao et al. [3]. developed quality-aware region selection subnet uses pseudo-labels to select high and low quality regions, highlighting the former and suppressing the latter in fusion. DIGRNet [11] proposed an interpretable mechanism called interference degree to assess the quality of depth maps and re-weight the contribution of single-modality features. DPANet [9] assesses depth quality by comparing depth maps to the ground truth following threshold segmentation, generating pseudo-labels that guide the depth data's utilization. D3Net [15] uses a depth depurator unit to coarsely compare the similarity between Sdepth and Srgbd to decide whether to use rgbnet or rgbdnet. However, this method does not fully account for the contribution differences across various regions of the depth map. Its performance heavily depends on branch selection, and it may fail when neither the RGB nor depth streams can adequately capture salient objects. In contrast, our approach allows for a more detailed evaluation of the contributions of different regions within each channel to saliency detection. And we uniquely balances the complementary nature of RGB and depth information, enabling efficient fusion without requiring extra labeling or learning processes. Beyond RGB-D SOD, recent multimodal studies have further shown that effective fusion often relies on explicit modeling of cross-modal complementarity, modality-specific interaction, and fine-grained alignment. For example, CONTRINet [45] improved multimodal saliency prediction by modeling complementary flows across modalities. Other works explored synergistic knowledge transfer across large multimodal models [44], modality-specific interaction analysis in vision-language systems [58], and implicit local alignment for fine-grained cross-modal matching [56]. Although these methods were developed for different multimodal tasks, they consistently suggest that different modalities may contribute unequally and should not be fused in a uniform manner. This observation is closely related to our motivation. However, existing methods do not explicitly estimate channel-wise contribution for RGB-D salient object detection, especially when the usefulness of depth cues varies across scenes and local regions.

To overcome these challenges, we propose a novel method that accurately perceives channel contributions, enabling the use of existing saliency detection tools without the need for additional labeling or training. This approach ensures a more reliable and efficient integration of depth data into salient object detection tasks.

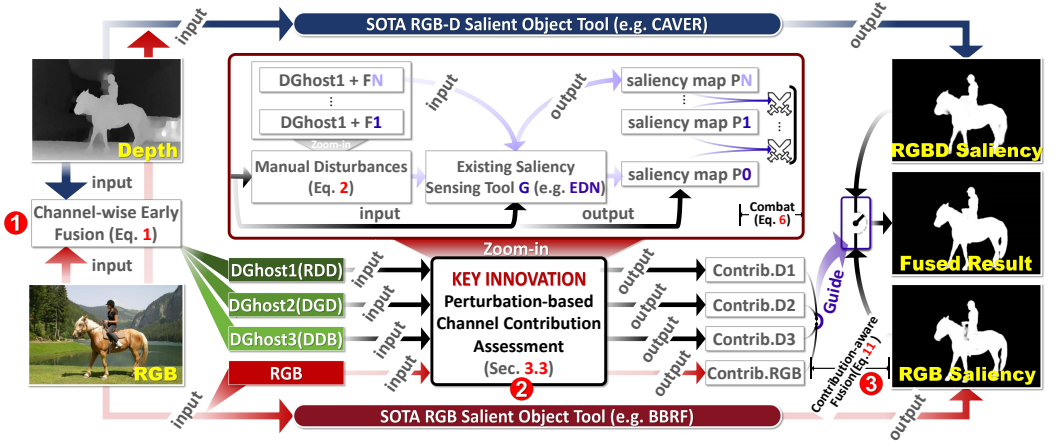


Fig. 3. The architecture of our channel-wise contribution assessment method comprises three main parts. Firstly, the channel-wise early fusion (indicated by ❶, Sec. 3.2) sequentially recombines the depth (D) channel with each of the three RGB channels to create new data termed “DGhosts”. Secondly, the DGhosts and RGB inputs are fed into the saliency perturbation-based channel contribution assessment (indicated by ❷, Sec. 3.3), which outputs RGB contribution score (Contrib.RGB) and DGhosts contributions (Contrib.D1, Contrib.D2, and Contrib.D3). Thirdly, guided by the contributions of each channel, the contribution-aware fusion (indicated by ❸) combines RGB saliency and RGB-D saliency to produce the final saliency result. Please refer to Sec. 3.4 for detailed technical details, motivations and advantages. The key advantage of this approach is its capability to accurately discern the contribution of each channel (❷) and to dynamically adapt the fusion process (❸) according to the specific contribution of each channel. This significantly boosts the precision and reliability of salient object detection.

3 Proposed Method

3.1 Method Overview

The core concept of our method is to perceive the contributions of both RGB and D channels and balance the complementary status between them for effective fusion. Illustrated in Fig. 3, our method, adopting a dual-stream approach for RGB and RGB-D salient object detection, respectively, featuring separate branches for RGB channel (red arrows) and RGB-D channels (blue arrows), is structured around three key components: 1) channel-wise early fusion, detailed in Sec. 3.2; 2) saliency perturbation-based channel contribution assessment, explained in Sec. 3.3; and 3) contribution-aware fusion, covered in Sec. 3.4.

Our method begins by combining a single-channel depth map with each channel of a three-channel RGB image (i.e., the R, G, or B channel), while duplicating the depth data to construct a set of three-channel inputs termed DGhosts (i.e., RDD, DGD, and DDB), as shown in the green rectangle of Fig. 3-❶. The key innovation of our approach, the saliency perturbation-based channel contribution assessment module (Fig. 3-❷), then evaluates both the DGhosts and the original RGB input to estimate their respective contribution scores. This step leverages the inherent robustness of saliency models to controlled perturbations (e.g., noise) to quantitatively assess the contribution of each channel. By introducing perturbations with progressively increasing intensity, we measure how each channel affects saliency prediction under input degradation. The final step is a contribution-aware fusion process (Fig. 3-❸), where the saliency result is produced by adaptively combining the contribution scores of the RGB and depth channels through a routing-style fusion scheme. This design allows the fusion process to adjust dynamically according to the estimated

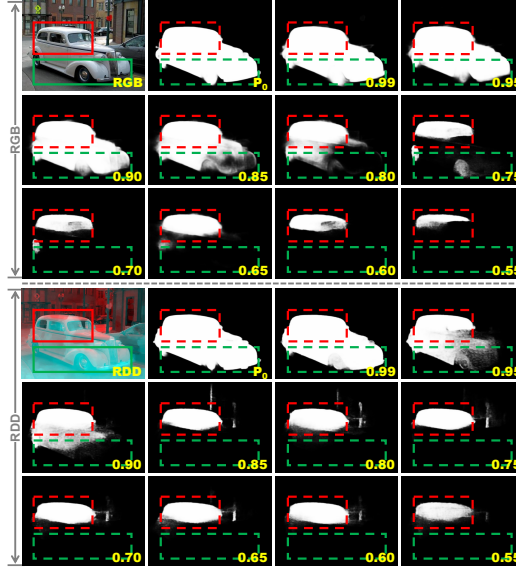


Fig. 4. The saliency changes in RGB and DGhost (e.g., RDD, Eq. 1). P_0 is the saliency map (generated by [52]) without noise disruptions. Saliency maps are affected by different levels of noise disruptions, with yellow numbers indicating the signal-to-noise ratio (lower numbers correspond to higher noise disruptions). High-contribution areas (highlighted by red boxes) exhibit greater saliency resilience to noise interference compared to low-contribution regions (indicated by green boxes). This means that after adding noise disruptions, the saliency of high-contribution areas changes slowly.

channel contributions, thereby generating more precise saliency maps by better exploiting the complementary strengths of RGB and depth information.

3.2 Channel-wise Early Fusion

To explore the inherent resilience of saliency models to noise disruptions and evaluate the robustness and significance of each channel, one direct approach is to individually introduce noise to the depth and RGB channels and observe the corresponding changes in their saliency outputs. However, this method comes with several limitations. First, applying noise solely to the depth channel offers a narrow view of the depth channel's contribution, neglecting how depth data interact with RGB channels. This singular focus fails to capture the complex, interdependent relationship between depth and RGB information; Second, the simplistic addition of noise to only the depth channel assumes a uniform impact across all scenes and objects, which is rarely the case. Different scenarios may exhibit varying degrees of sensitivity to depth information quality, leading to inaccurate assessments of channel contributions. To tackle these issues, we introduce a straightforward yet effective method called channel-wise early fusion. As depicted in Fig. 3-①, this technique involves substituting two RGB channels with depth information² to create a new form of data named DGhost (highlighted in the green rectangle). The transformation is defined as:

$$\text{DGhost}_i = \{\text{RGB}_i, \text{D}, \text{D}\}, \quad (1)$$

where RGB_i , where $i \in 1, 2, 3$, represents the individual R, G, or B channel within the three-channel RGB. Similarly, DGhost_i , with $i \in 1, 2, 3$, corresponds to RDD, DGD, and DDB, respectively. By creating D Ghosts with different combinations of RGB and depth channels (e.g., RDD, DGD, DDB),

²Ablation experiments are shown in Table. 5.

we establish a framework where the impact of noise disruptions can be analyzed more effectively. This approach allows us to observe how each channel responds to noise disruptions and how their contributions change accordingly (see Sec. 3.3). Without this comparative analysis against RGB channels, the assessment of depth channel contribution would lack a meaningful reference point for evaluation. Next, we will detail how to evaluate channel contributions.

3.3 Perturbation-based Channel Contribution Assessment

Existing fusion-based RGB-D salient object detection methods³ [4, 16, 51, 64, 66] incorporate depth maps by optimizing depth quality and assessing depth map reliability to mitigate potential negative impacts. For example, the poor quality of depth data can lead to inaccurate saliency results due to incorrect spatial cues in the depth map. This approach can reduce the fusion of incorrect spatial cues in the depth map, thereby minimizing their interference. However, they do not fully account for the varying contribution of different channels, since channels vary in contribution depending on the scenes, leading to potential perceptual biases and the risk of generating misleading information.

As mentioned-above, Fig. 2 illustrates that high/low-contribution regions within a depth map perform diversely. Extensive experimental results presented in Fig. 4 indicate that both RGB and DGhost channels are susceptible to noise disruptions, but channels with varying contributions exhibit different levels of resilience to noise disruptions. For instance, when equal noise disruption levels (quantified by signal-to-noise ratio) are introduced, high-contribution areas (highlighted by red boxes) exhibit greater resilience to noise disruptions compared to low-contribution regions (indicated by green boxes). In light of this, we pose a question: Is it possible to detailedly analyze contributions of the RGB channel and depth channel, especially the different regions within the depth channel to assist the final saliency map generation? To further verify the feasibility of this idea, we plan to add noise disruptions to different channels to see the saliency changes.

Building on this insight, we propose a saliency perturbation-based channel contribution assessment method, illustrated in Fig. 3-2. The core idea is to introduce controlled perturbations into the input channel data and assess the contributions of different channels by measuring the resulting saliency variations. When a channel reaches its “failure” or limit condition, it signifies that the noise level has surpassed the channel’s capacity to contribute meaningful information for saliency detection. This threshold is crucial for understanding the channel’s contribution: it provides a quantifiable point where the channel’s usefulness changes from beneficial to negligible. By systematically introducing noise and observing when the saliency detection performance begins to decline significantly, we can infer the robustness and, implicitly, the contribution of each channel.

To implement saliency perturbation-based channel contribution assessment, we take the D Ghosts from the channel-wise early fusion (Sec. 3.2) and RGB as input (i.e., the black and red arrows in Fig. 3-2), and obtain the corresponding channel contribution scores. In particular, focusing on the zoomed-in area of Fig. 3-2, we apply a series of controlled perturbations to clean images (i.e., $DGhost_i$, $i \in \{1, 2, 3\}$, or RGB), including salt-and-pepper noise, Gaussian blur, varying illumination, and compression. Among these perturbations, salt-and-pepper noise consists of random bright and dark points, which can affect image regions in a smaller-scale or more localized manner, resulting in subtle saliency changes. Accordingly, following the signal-to-noise ratio principle, we introduce perturbations at different levels with consistent residuals to probe channel contributions. For channel contribution assessment, we propose two methods: coarse-grained and fine-grained. We elaborate on them in the following sections.

3.3.1 Coarse-grained Channel Contribution Assessment. Starting with a clean image x , either a $DGhost_i$ or an RGB input, we apply the t -th perturbation x_{noi}^t to generate a perturbed image. This

³Fusion-based RGB-D SOD refers to methods that combine RGB and depth data to enhance salient object detection accuracy.

process is mathematically represented as:

$$F_t = x + x_{noi}^t, \quad (2)$$

where $F_t \in \mathbb{R}^{H \times W \times 3}$ ($t \in [1, N]$) denotes the perturbed image obtained by applying the t -th perturbation level (e.g., salt-and-pepper noise), and N is the maximum perturbation level⁴. Here, H and W denote the height and width of the image, respectively. The noisy image F_t is then processed by a saliency detection tool G ⁵ (e.g., [52], [65], and [31]) to produce a saliency map $P_t \in \mathbb{R}^{H \times W \times 1}$:

$$P_t = G(F_t). \quad (3)$$

here, G serves as a saliency probing function that characterizes the relative response of the input under progressively increasing perturbations. Accordingly, the contribution estimation is based on the variation trend of saliency maps across perturbation levels, rather than the absolute prediction quality of an individual DGhost input.

When a channel image is fully burdened with noise disruptions, its corresponding saliency map, P_t , starts to exhibit a range of issues: detail loss, deformity in salient objects, fuzzy edges, and mistakes in pinpointing locations. This effect is illustrated in Fig. 4, where regions of high contribution demonstrate a robust defense against disturbances, contrasting sharply with the more affected low-contribution areas.

To measure the impact of noise disruptions at the t -th level on different channels, intuitively, we can directly assess each channel's contribution based on the magnitude of saliency map changes; however, this may not accurately reflect contribution, as scene variations can be significant. Thus, we introduce to perceive contribution in a relative change manner by comparing the noisy image's saliency map P_t , with the original clean channel x 's P_0 , using saliency evaluation metrics like the S-measure[13] for evaluating saliency changes.

We take DGhost₁ as an example to illustrate the whole process in detail. As shown in Fig. 3-2, when DGhost₁ is processed by the saliency perturbation-based channel contribution assessment module, its contribution score, denoted as Contrib.D1, is determined according to the saliency variation sequence under progressively increasing perturbation levels.

Given the saliency map P_0 of the original input and the saliency map P_i obtained after applying the i -th perturbation level, we first define the saliency similarity sequence as:

$$S_i = \text{Sm}(P_0, P_i), \quad i \in [0, N], \quad (4)$$

where $\text{Sm}(\cdot, \cdot)$ denotes the S-measure for measuring saliency similarity between two saliency maps.

Then, the saliency variation between two consecutive perturbation levels is defined as:

$$\Delta_i = S_{i-1} - S_i, \quad i \in [1, N], \quad (5)$$

where Δ_i measures saliency similarity decrease as the perturbation level increases from $i - 1$ to i .

Based on the variation sequence $\{\Delta_1, \Delta_2, \dots, \Delta_N\}$, the contribution score of DGhost₁ is given by:

$$\text{Contrib.D1} = T_C(\Delta_1, \Delta_2, \dots, \Delta_N), \quad (6)$$

where $T_C(\cdot)$ identifies the first perturbation level at which the saliency variation exceeds a dynamic threshold and returns it as the contribution score. Following the empirical setting, the threshold is defined as half of the range between the maximum and minimum values of the saliency similarity sequence. The contribution scores of DGhost₂, DGhost₃, and RGB are computed similarly.

Fig. 3-2 demonstrates that, using the method described, we can determine the contribution of each data channel without the need for additional labeling or learning.

⁴For the RGB channels, the maximum value of N is 13, whereas for the D channel, the maximum value of N is 30.

⁵Any SOTA SOD method is applicable in this process. Detailed ablation studies are provided in Sec. 4.4.1.

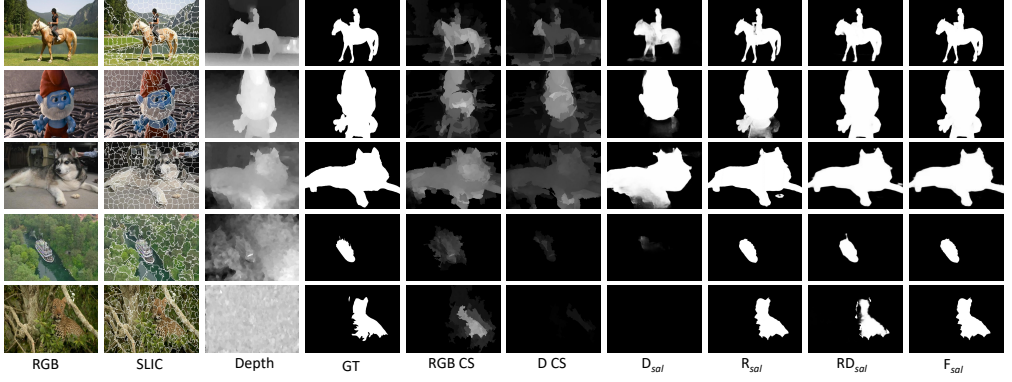


Fig. 5. Visual representations of the results include SLIC, RGB channel contribution score map (RGB CS), depth channel contribution score map (D CS), depth saliency map (D_{sal}), RGB saliency map (R_{sal}), RGB-D saliency map (RD_{sal}), and the ultimate saliency map (F_{sal}) in our proposed approach.

This allows us to calculate the contribution scores for RGB (Contrib.RGB) and $DGhost_i$ (Contrib.D1, Contrib.D2, and Contrib.D3). The overall contribution score for the D channel is then derived through a process of averaging the scores from Contrib. D1, Contrib.D2, and Contrib.D3:

$$D = \text{AVG}(\text{Contrib.D1}, \text{Contrib.D2}, \text{Contrib.D3}), \quad (7)$$

where $\text{AVG}(\cdot)$ represents the operation of averaging. Note that other contribution score collection methods should also be feasible, such as maximization, but maximization has a slightly inferior performance. Here, we simply use AVG for simplicity. Contrib. D1, Contrib.D2, and Contrib.D3 refer to the contribution scores associated with $DGhost_1$, $DGhost_2$, and $DGhost_3$, respectively.

3.3.2 Fine-grained Channel Contribution Assessment. Eq. 4 enables us to calculate the overall contributions of RGB and D channels. Yet, the influence of the surrounding environment on salient objects means that contributions can vary across different regions, as shown in the example in Fig. 2. Merely fusing contributions from the whole image of the input channel often leads to less than optimal outcomes due to this rough perceptual assessment.

To address this issue, we introduce a fine-grained method for assessing channel contributions that calculates the contribution of each region within the channels. We employ the Simple Linear Iterative Clustering [2] (SLIC), to segment the channels into distinct regions with superpixels. Visualizations of the results of SLIC can be seen in Fig. 5-SLIC. However, calculating contribution scores on a region-by-region basis (as per equations from Eq. 2 to Eq. 4) is time-consuming. Thus, we attempt to accelerate the contribution score calculation process at two levels: 1) We dynamically adjust the division of super-pixel sizes according to the size of salient objects; 2) From Eq. 4, we derive an approximate resilience noise disruption level t for the entire image. This value guides us for refining super-pixel regions through a forward and backward search, thereby reducing computational overhead. For example, if our analysis yields a t of 0.7 for an image, we search around this value, incrementally adding noise disruption levels like 0.6, 0.65, 0.7, and 0.75. This method allows us to avoid starting anew and instead adjusts noise disruption levels incrementally from a signal-to-noise ratio of 1 forward.

To implement fine-grained channel contribution assessment, we further partition the image into M superpixel regions, denoted as $\{sp_j\}_{j=1}^M$, where each region contains spatially adjacent pixels with similar characteristics. Based on the image-level formulation in Eqs. 4–6, we extend the assessment to the superpixel level for more precise local analysis. Still taking $DGhost_1$ as an

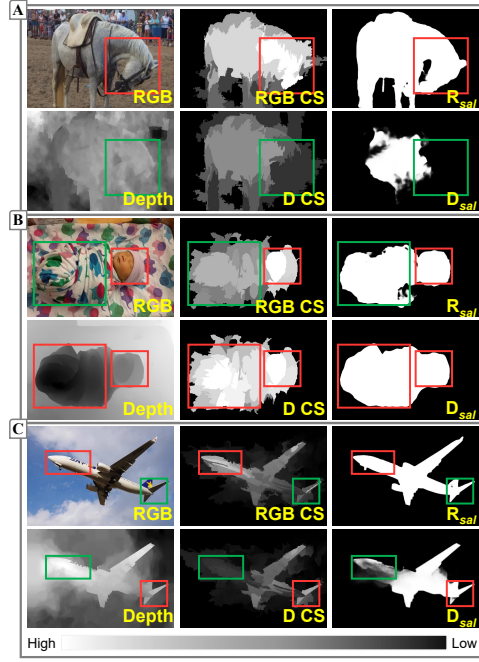


Fig. 6. The highlighted regions within the score maps indicate regions with higher contributions. For subfigure A, the pronounced brightness in the RGB channels' score map (RGB CS) denotes its higher contribution on the final saliency map, whereas, in subfigure B, it is the D channel's score map (D CS), distinguished by its brightness, that notably contributes to the ultimate saliency map. In subfigure C, fine-grained regions within the D channel with higher contributions can produce more accurate saliency maps. R_{sal} and D_{sal} separately denote RGB saliency map and D saliency map. High/low-contribution regions are highlighted by red/green boxes, respectively.

example, the regional saliency similarity sequence in region sp_j is defined as:

$$S_i(sp_j) = \text{Sm}(P_0(sp_j), P_i(sp_j)), \quad i \in [0, N], \quad (8)$$

where $P_0(sp_j)$ and $P_i(sp_j)$ denote the saliency values within superpixel region sp_j before and after the i -th perturbation, respectively.

The regional saliency variation is then computed by

$$\Delta_i(sp_j) = S_{i-1}(sp_j) - S_i(sp_j), \quad i \in [1, N], \quad (9)$$

where $\Delta_i(sp_j)$ reflects saliency similarity decrease in region sp_j as the perturbation level increases.

Accordingly, the fine-grained contribution score of DGhost₁ on sp_j is given by

$$\text{Contrib.D1}(sp_j) = T_C(\Delta_1(sp_j), \Delta_2(sp_j), \dots, \Delta_N(sp_j)). \quad (10)$$

Compared with the image-level formulation, this fine-grained strategy enables more accurate estimation of local channel contributions, thereby supporting more precise RGB-depth fusion. The effectiveness of this refinement is validated in Table 4, and visual examples are shown in Fig. 6.

3.4 Contribution-aware Fusion

While numerous fusion methods exist, this paper does not aim to explore new fusion techniques. Instead, the focus lies in integrating the perceived contributions into the fusion process. To ensure clarity and maintain focus on the main theme of this paper, we adopt the simplest fusion method

as a baseline and modify it accordingly. However, it's worth noting that our approach can also be applied to other fusion methods, such as cross-modal fusion [46], by incorporating contribution scores during attention computation.

Building upon the saliency perturbation-based channel contribution assessment (see Sec. 3.3), we derive contribution scores for the RGB and D channels. Guided by these scores, we introduce a contribution-aware fusion scheme that adopts a binary routing-style strategy to combine the channels. This design aims to reduce interference from low-contribution or unreliable depth information, thereby improving modality complementarity. The contribution-aware fusion scheme (depicted in Fig. 3-) comprises two branches: one uses the RGB salient object detector [31] to produce RGB saliency (illustrated by the red arrows in Fig. 3), while the other uses the RGB-D salient object detector [34] to generate RGB-D saliency (depicted by the blue arrows in Fig. 3).

Our contribution-aware fusion entails a direct selection of RGB saliency or RGB-D saliency based on each channel's contribution, in order to mitigate the impact of low-contribution, no contribution, or negatively influential D regions. Specifically, when D exhibits low-contribution (e.g., Fig. 1-B), leveraging RGB-D fusion may inadvertently introduce negative effects; thus, it is preferable to discard D and directly utilize RGB saliency (R_{sal}). Conversely, when the D channel exhibits high-contribution and the corresponding RGB channel shows low-contribution (e.g., Fig. 1-A), we opt for RGB-D saliency (RD_{sal}), as the high-contribution D provides valuable spatial information for challenging RGB samples, resulting in an improved saliency map compared to RGB saliency. By leveraging RGB and D channel contribution scores to guide the binary routing fusion, we generate the fused saliency (F_{sal}) as follows:

$$F_{sal} = (1 - \omega_g) \times R_{sal} + \omega_g \times RD_{sal}, \quad (11)$$

where $\omega_g \in \{0, 1\}$ is a global routing weight. It is determined by the global contribution scores of the RGB and D channels:

$$\omega_g = \begin{cases} 1, & \text{if Contrib.RGB} < \text{Contrib.D}, \\ 0, & \text{otherwise.} \end{cases} \quad (12)$$

where, Contrib.RGB denotes the global contribution score of the RGB channel, and Contrib.D denotes the global contribution score of the depth channel.

Likewise, for the fine-grained channel contribution assessment based on SLIC superpixel segmentation (Sec. 3.3.2), the fusion is performed at the region level:

$$F_{sal}(sp_j) = (1 - \omega_j) \times R_{sal}(sp_j) + (\omega_j) \times RD_{sal}(sp_j). \quad (13)$$

Our method assesses channel contributions and performs routing fusion seamlessly without the need for extra labeling or learning processes. We provide visual examples of the contributions from RGB and D channels, as well as the saliency maps from RGB (R_{sal}), RGB-D (RD_{sal}), and the fused result (F_{sal}), illustrated in Fig. 5. This approach significantly minimizes the impact of regions in the D channel that have low or negative contributions, ensuring a more harmonious integration of RGB and D channels.

4 Experiment

4.1 Implementation Details and Evaluation Metrics

Our experiments were performed on a workstation equipped with an NVIDIA GeForce RTX 3090 GPU (with 24GB of memory) and 16GB RAM. Our method was implemented using Python 3.8 with PyTorch 1.9.

We conducted experiments on nine publicly available RGB-D SOD datasets: NJU2K [22], NLPR [36], SIP [15], STEREO [33], SSD [73], LFSD [27], DUT [37], ReDWeb-S [55] and COME15K-E [60]. For

Table 1. Quantitative evaluation of key component. “BL1” is the baseline RGB saliency model, e.g., BBRF [31], and “BL2” is the baseline RGB-D saliency model, e.g., CAVER [34]. “CeF”: channel-wise early fusion; “RGB” and “D”: the RGB and D channels; “CG”: coarse-grained contribution assessment; “FG”: fine-grained contribution assessment; “RF”: contribution-aware fusion. $\uparrow$ denotes larger is better, and $\downarrow$ denotes smaller is better. The best results are highlighted in bold.

NO.	BL1	BL2	CeF	RGB	D	CG	FG	RF	NJU2K			NLPR			SIP			SSD			LFSD		
									$S_m \uparrow$	$F_\beta \uparrow$	MAE $\downarrow$	$S_m \uparrow$	$F_\beta \uparrow$	MAE $\downarrow$	$S_m \uparrow$	$F_\beta \uparrow$	MAE $\downarrow$	$S_m \uparrow$	$F_\beta \uparrow$	MAE $\downarrow$	$S_m \uparrow$	$F_\beta \uparrow$	MAE $\downarrow$
1	✓								0.908	0.892	0.034	0.913	0.880	0.028	0.915	0.901	0.028	0.870	0.829	0.041	0.856	0.825	0.077
2		✓							0.923	0.909	0.028	0.930	0.898	0.023	0.895	0.891	0.043	0.887	0.844	0.038	0.882	0.854	0.056
3	✓	✓		✓	✓	✓	✓	✓	0.926	0.910	0.026	0.932	0.902	0.024	0.901	0.895	0.040	0.888	0.847	0.035	0.887	0.861	0.052
4	✓	✓	✓		✓	✓	✓	✓	0.926	0.913	0.027	0.933	0.906	0.023	0.900	0.889	0.040	0.893	0.850	0.036	0.886	0.859	0.052
5	✓	✓	✓	✓	✓	✓	✓	✓	0.929	0.920	0.025	0.937	0.910	0.020	0.902	0.893	0.038	0.895	0.854	0.036	0.889	0.864	0.050
6	✓	✓	✓	✓	✓	✓	✓	✓	0.925	0.907	0.029	0.933	0.904	0.020	0.898	0.875	0.041	0.885	0.841	0.035	0.873	0.856	0.057
7	✓	✓	✓	✓	✓	✓	✓	✓	0.931	0.920	0.024	0.936	0.912	0.018	0.909	0.890	0.034	0.894	0.863	0.035	0.892	0.865	0.051
8	✓	✓	✓	✓	✓	✓	✓	✓	0.926	0.913	0.030	0.936	0.909	0.022	0.906	0.890	0.035	0.893	0.861	0.036	0.890	0.868	0.049
9	✓	✓	✓	✓	✓	✓	✓	✓	0.936	0.927	0.022	0.940	0.913	0.017	0.917	0.902	0.028	0.901	0.869	0.032	0.898	0.875	0.046

a fair comparison with other RGB-D SOD methods, we use the same training/testing data split scheme and employ the same training parameters and strategies as in [12, 34].

For perturbation-based channel contribution assessment, we apply controlled degradations to the clean RGB image or DGhost input, including salt-and-pepper noise, Gaussian blur, illumination variation, and compression distortion. Unless otherwise specified, salt-and-pepper noise is used by default due to its better probing stability. Following Sec. 3.3, the maximum perturbation level is set to $N = 13$ for RGB and $N = 30$ for D. For each level, we compute the saliency similarity sequence $\{S_i\}_{i=0}^N$ with respect to the clean saliency map, and then obtain the saliency variation sequence by $\Delta_i = S_{i-1} - S_i$. The operation $T_C(\cdot)$ returns the first perturbation level at which Δ_i exceeds a dynamic threshold, defined as half of the range of the saliency similarity sequence.

We use five widely adopted metrics for quantitative evaluation, including S-measure [13] (S_m), F-measure [1] (F_β , $\beta^2 = 0.3$), weighted F-measure [32] (F_β^ω), E-measure [14] (E_m), and MAE [61]. Specifically, S_m evaluates structural similarity, F_β measures regional precision-recall balance, F_β^ω improves F_β with weighted precision and recall, E_m captures both global and local consistency, and MAE measures the average pixel-wise absolute error.

4.2 Component Evaluation

To validate the effectiveness of our method, we have conducted an extensive component evaluation, and the results have been shown in Table 1. We treat the original RGB (e.g., BBRF, line 1) or RGB-D (e.g., CAVER, line 2) SOD model as baselines.

By comparing line 3 and line 9, we can easily validate the effectiveness of the proposed channel-wise early fusion (CeF, Sec. 3.2). In this scenario, when CeF is removed and only the three-channel depth is used instead of DGhost (Eq. 1), the model performs worse than when equipped with CeF, e.g., the S_m metric has been decreased from 0.898 to 0.887 in the LFSD set. This highlights the importance of incorporating the RGB channel in the depth channel, as without this comparative analysis against the RGB channels, evaluating the contribution of the depth channel would lack a meaningful reference point.

As highlighted in lines 4-5, solely evaluating either the RGB or D channel independently is less effective compared to simultaneous assessment of both RGB and D channels (line 9), e.g., the S_m metric has been decreased from 0.898 to 0.886 to 0.889, respectively, in the LFSD set. This observation stems from the inherent complementarity between RGB and D channels, underscoring our proposition that optimal synergy is attained by jointly considering the contributions of RGB and D channels.



Fig. 7. Visual comparisons between our method and several representative SOTA RGB-D SOD methods under different challenging scenes.

To validate the effectiveness of the saliency perturbation-based channel contribution assessment described in Sec. 3.3, we conducted experiments by replacing it with a basic alternative (described in line 6). Specifically, we first roughly locate the salient object through RGB saliency analysis, and then employ the Segment Anything model (SAM [23]) to segment the object and obtain the Pseudo Ground Truth (PGT). Subsequently, we independently measure the structural similarity between the RGB saliency and PGT, and between the depth saliency and PGT. A lower similarity for depth saliency indicates lower quality, i.e., lower contribution, in which case RGB saliency is adopted. Conversely, a higher similarity suggests that the RGB-D salient object detector is more suitable for achieving better accuracy. Compared to this basic method, our approach — coarse-grained channel contribution assessment (outlined in line 7, Sec. 3.3.1) — demonstrates a significant performance enhancement. For instance, it yielded an average increase in the Sm metric by 0.7%, indicating a superior ability to discern salient features. Moreover, the refinement to a fine-grained contribution assessment strategy (referenced in line 9, Sec. 3.3.2) further amplified this effect, elevating the Sm metric from 0.931 to 0.936 on the NJU2K dataset. This improvement aligns with our hypothesis that image regions are influenced variably by their surroundings, leading to diverse contributions across different regions.

Results in the Table 1 also demonstrate that our method (line 9) outperforms the naive bi-linear fusion (line 8). This is attributed to our routing-style contribution-aware fusion, which effectively mitigates the impact of low-contribution, no contribution, or negatively influential D regions. In contrast, the naive bi-linear fusion combines the two types of saliency maps without considering the contribution of corresponding channels, leading to suboptimal results.

4.3 Comparison with Other SOTA Methods

To demonstrate the effectiveness of our method, we compare it with 15 state-of-the-art (SOTA) RGB-D SOD methods, including HRTrans [43], CAVER [34], PICRNet [12], LAFB [49], EMTrans [7], UniTR [18], CPNet [20], SACNet [48], MAGNet [67], CATNet [42], HENet [17], PLFRNet [19],

Table 2. Comparison with state-of-the-art RGB-D SOD methods on 9 benchmark datasets. OursI is the results of our method using coarse-grained channel contribution assessment (Sec. 3.3.1), and OursII is the results of the fine-grained channel contribution assessment (Sec. 3.3.2). The best two results are highlighted using **Red** and **Blue** in the order.

Dataset	Metric	HRTrans ₂₃	CAVER ₂₃	PICRNet ₂₃	LAFB ₂₄	EMTrans ₂₄	UniTR ₂₄	CPNet ₂₄	SACNet ₂₄	MAGNet ₂₄	CATNet ₂₄	HENet ₂₅	PLFRNet ₂₅	ACINet ₂₅	BTNet ₂₅	LESOD ₂₆	OursI (CAVER)	OursII (CAVER)
Params (M)		58.9	93.7	111.9	139.7	--	146.6	216.5	327.7	16.1	262.6	10.7	41.9	85.7	13.8	2.9	189.9	189.9
FLOPs (G)		17.3	63.9	27.0	453.0	--	135.0	129.3	--	9.9	341.8	11.9	25.1	--	18.5	3.3	85.3	85.3
Speed (FPS)		21.1	27.7	25.9	21.6	--	9.7	13.3	--	34.1	--	80.9	32.6	--	--	88.7	15.1	11.3
RGB backbone		HRFormer	RN50	Swin T	RN50	PVTv2	Swin T	Swin T	Swin T	SMT	Swin T	MNetV1	PVTv2	P2T	SMT	EdgeNext	RN50	RN50
Depth backbone		RN18	RN50			PVTv2	Swin T	Swin T	Swin T	MNetV2	Swin T	MNetV1	PVTv2	P2T	DepthNet	MNetV3	RN50	RN50
NIJ2K	$S_m \uparrow$	0.928	0.923	0.926	0.924	0.930	0.925	0.934	0.935	0.928	0.932	0.929	0.929	0.934	0.934	0.920	0.931	0.936
	$F_\beta \uparrow$	0.920	0.920	0.919	0.918	0.926	0.931	0.930	0.932	0.935	0.937	0.930	0.934	0.939	--	0.921	0.931	0.936
	$F_{\beta_2} \uparrow$	0.911	0.909	0.909	0.904	0.915	0.918	0.923	0.925	0.912	0.917	0.905	0.914	0.919	--	0.894	0.920	0.927
	$E_m \uparrow$	0.928	0.931	0.933	0.924	0.934	0.939	0.934	0.939	0.962	0.960	0.957	0.960	0.962	0.958	0.952	0.932	0.934
	MAE $\downarrow$	0.029	0.028	0.029	0.029	0.027	0.025	0.025	0.024	0.027	0.026	0.029	0.028	0.025	0.025	0.033	0.024	0.022
NLPR	$S_m \uparrow$	0.940	0.930	0.935	0.929	0.939	0.930	0.940	0.936	0.935	0.941	0.942	0.934	0.941	0.943	0.926	0.936	0.940
	$F_\beta \uparrow$	0.917	0.903	0.911	0.901	0.916	0.922	0.922	0.917	0.926	0.933	0.935	0.929	0.934	--	0.915	0.915	0.918
	$F_{\beta_2} \uparrow$	0.914	0.898	0.906	0.895	0.912	0.911	0.916	0.910	0.903	0.912	0.912	0.909	0.912	--	0.889	0.912	0.914
	$E_m \uparrow$	0.967	0.961	0.966	0.958	0.966	0.970	0.970	0.964	0.968	0.971	0.972	0.970	0.972	0.970	0.962	0.968	0.968
	MAE $\downarrow$	0.018	0.023	0.019	0.021	0.017	0.018	0.018	0.020	0.018	0.018	0.017	0.018	0.016	0.016	0.021	0.018	0.017
SIP	$S_m \uparrow$	0.888	0.895	0.899	0.897	0.904	0.896	0.907	0.888	0.888	0.910	0.906	0.882	0.910	0.918	0.855	0.909	0.917
	$F_\beta \uparrow$	0.890	0.897	0.900	0.903	0.912	0.915	0.917	0.908	0.906	0.928	0.917	0.896	0.929	--	0.871	0.908	0.915
	$F_{\beta_2} \uparrow$	0.862	0.891	0.876	0.875	0.866	0.888	0.895	0.880	0.884	0.901	0.886	0.871	0.910	--	0.844	0.890	0.902
	$E_m \uparrow$	0.920	0.935	0.934	0.937	0.938	0.940	0.941	0.934	0.938	0.951	0.947	0.933	0.949	0.951	0.915	0.949	0.952
	MAE $\downarrow$	0.046	0.043	0.040	0.041	0.039	0.037	0.035	0.039	0.044	0.035	0.038	0.047	0.034	0.031	0.059	0.034	0.028
STERE	$S_m \uparrow$	0.912	0.914	0.916	0.906	0.925	0.905	0.920	0.917	0.921	0.921	0.921	0.919	0.922	0.928	0.908	0.921	0.928
	$F_\beta \uparrow$	0.894	0.882	0.904	0.896	0.909	0.901	0.907	0.905	0.920	0.919	0.921	0.918	0.923	--	0.905	0.910	0.909
	$F_{\beta_2} \uparrow$	0.886	0.888	0.898	0.882	0.905	0.890	0.902	0.901	0.886	0.884	0.890	0.885	0.886	--	0.872	0.900	0.911
	$E_m \uparrow$	0.930	0.936	0.937	0.930	0.937	0.932	0.934	0.929	0.957	0.955	0.956	0.956	0.959	0.955	0.949	0.937	0.944
	MAE $\downarrow$	0.033	0.033	0.036	0.037	0.031	0.033	0.029	0.030	0.030	0.030	0.031	0.030	0.030	0.027	0.035	0.030	0.027
SSD	$S_m \uparrow$	0.868	0.887	0.880	0.880	0.886	0.879	0.887	0.896	0.884	--	0.893	0.886	--	--	0.880	0.894	0.901
	$F_\beta \uparrow$	0.835	0.869	0.853	0.865	0.862	0.873	0.868	0.884	0.878	--	0.881	0.881	--	--	0.870	0.859	0.863
	$F_{\beta_2} \uparrow$	0.821	0.844	0.840	0.839	0.846	0.857	0.858	0.869	0.829	--	0.842	0.841	--	--	0.825	0.863	0.869
	$E_m \uparrow$	0.910	0.919	0.917	0.926	0.922	0.923	0.920	0.929	0.932	--	0.934	0.936	--	--	0.922	0.924	0.932
	MAE $\downarrow$	0.045	0.038	0.045	0.040	0.039	0.036	0.038	0.032	0.042	--	0.037	0.040	--	--	0.048	0.035	0.032
LFSD	$S_m \uparrow$	0.887	0.882	0.888	0.864	--	0.873	0.892	--	0.888	0.894	0.898	0.890	0.896	0.898	0.891	0.892	0.898
	$F_\beta \uparrow$	0.883	0.878	0.889	0.862	--	0.874	0.890	--	0.891	0.891	0.898	0.889	0.899	--	0.891	0.871	0.893
	$F_{\beta_2} \uparrow$	0.858	0.854	0.862	0.829	--	0.851	0.868	--	0.849	0.849	0.848	0.855	0.868	--	0.851	0.865	0.875
	$E_m \uparrow$	0.912	0.904	0.917	0.888	--	0.896	0.920	--	0.923	0.927	0.933	0.925	0.931	0.927	0.922	0.901	0.915
	MAE $\downarrow$	0.055	0.056	0.052	0.065	--	0.057	0.049	--	0.053	0.051	0.049	0.051	0.047	0.047	0.053	0.051	0.046
DUT	$S_m \uparrow$	0.948	0.937	0.941	0.931	--	0.939	0.948	0.945	0.943	0.953	0.942	0.950	0.947	0.952	0.931	0.941	0.945
	$F_\beta \uparrow$	0.950	0.936	0.940	0.929	--	0.938	0.950	0.949	0.955	0.960	0.953	0.959	0.958	--	0.940	0.942	0.944
	$F_{\beta_2} \uparrow$	0.942	0.926	0.933	0.915	--	0.925	0.947	0.940	0.935	0.945	0.930	0.944	0.945	--	0.915	0.937	0.935
	$E_m \uparrow$	0.970	0.962	0.967	0.957	--	0.959	0.970	0.967	0.972	0.977	0.973	0.976	0.973	0.973	0.962	0.963	0.971
	MAE $\downarrow$	0.021	0.026	0.022	0.027	--	0.027	0.021	0.021	0.021	0.020	0.023	0.018	0.021	0.018	0.027	0.023	0.020
ReDWeb-S	$S_m \uparrow$	0.715	0.736	0.693	0.728	--	0.704	0.752	--	0.747	--	0.756	0.735	--	--	0.697	0.771	0.784
	$F_\beta \uparrow$	0.687	0.725	0.660	0.727	--	0.704	0.754	--	0.754	--	0.761	0.738	--	--	0.701	0.766	0.782
	$F_{\beta_2} \uparrow$	0.633	0.672	0.611	0.664	--	0.639	0.697	--	0.727	--	0.729	0.710	--	--	0.665	0.721	0.742
	$E_m \uparrow$	0.745	0.755	0.737	0.757	--	0.763	0.775	--	0.818	--	0.824	0.810	--	--	0.782	0.781	0.793
	MAE $\downarrow$	0.137	0.118	0.151	0.128	--	0.125	0.112	--	0.119	--	0.117	0.123	--	--	0.143	0.098	0.092
COME15K-E	$S_m \uparrow$	0.855	0.870	0.850	--	--	0.852	0.877	--	0.882	--	0.880	--	--	--	0.840	0.878	0.889
	$F_\beta \uparrow$	0.839	0.861	0.828	--	--	0.857	0.874	--	0.886	--	0.880	--	--	--	0.838	0.870	0.882
	$F_{\beta_2} \uparrow$	0.812	0.837	0.803	--	--	0.823	0.850	--	0.854	--	0.937	--	--	--	0.802	0.848	0.863
	$E_m \uparrow$	0.856	0.869	0.856	--	--	0.874	0.879	--	0.924	--	0.919	--	--	--	0.888	0.875	0.884
	MAE $\downarrow$	0.059	0.051	0.062	--	--	0.051	0.048	--	0.048	--	0.051	--	--	--	0.068	0.047	0.041

ACINet [41], BTNet [38], and LESOD [68]. For a fair comparison, Table 2 reports not only quantitative performance but also model complexity (Params, FLOPs), inference speed (FPS), and the corresponding RGB and depth backbone architectures, all evaluated under a unified setting.

4.3.1 Quantitative Comparison. Table 2 presents the quantitative evaluation results of our proposed method across nine benchmark datasets, with BBRF [31] and CAVER [34] serving as the baseline RGB SOD and RGB-D SOD models, respectively. “OursI” represents the outcomes of our method at the coarse-grained contribution assessment (Sec. 3.3.1), while “OursII” denotes the results of fine-grained contribution assessment (Sec. 3.3.2). Our method exhibits significant performance enhancements across all nine datasets, with the fine-grained assessment results surpassing the coarse-grained ones. For instance, leveraging CAVER [34], “OursI” achieves an average improvement of 2.1% and 1.8% over the baseline model across all nine challenging datasets concerning the S_m and F_β metrics, respectively. Furthermore, “OursII” demonstrates an average improvement of 0.5% and 0.7% over “OursI” across all nine challenging datasets regarding the S_m and F_β metrics.

Table 3. Ablation analysis of different combinations of SOTA RGB (e.g., EDN [52], BBRF [31]) and RGB-D (e.g., HENet [17] and LESOD [68]) Salient Object Tools. “OursI” is coarse-grained channel contribution assessment (Sec. 3.3.1), and “OursII” is fine-grained channel contribution assessment (Sec. 3.3.2).

Dataset		Complexity		NJU2K		NLPR		LFSD	
Metric		Params	FLOPS	$F_{\omega_b} \uparrow$	MAE $\downarrow$	$F_{\omega_b} \uparrow$	MAE $\downarrow$	$F_{\omega_b} \uparrow$	MAE $\downarrow$
RGB	EDN	21.8	25.8	0.857	0.049	0.844	0.034	0.806	0.086
	BBRF	74.4	27.9	0.892	0.034	0.880	0.028	0.825	0.077
RGB-D	CAVER	93.7	31.6	0.909	0.028	0.898	0.023	0.854	0.056
	PICRNet	111.9	27.0	0.909	0.029	0.906	0.019	0.862	0.052
	HENet	10.7	11.9	0.930	0.029	0.935	0.017	0.898	0.049
	LESOD	2.9	3.3	0.921	0.033	0.915	0.021	0.891	0.053
OursI	CAVER + EDN	137.3	83.2	0.916	0.025	0.900	0.019	0.862	0.053
	PICRNet + EDN	155.5	78.6	0.911	0.027	0.904	0.019	0.866	0.053
	HENet + EDN	54.3	63.5	0.929	0.031	0.932	0.018	0.899	0.048
	LESOD + EDN	46.5	54.9	0.923	0.033	0.912	0.023	0.893	0.051
	CAVER + BBRF	189.9	85.3	0.920	0.024	0.912	0.018	0.865	0.051
	PICRNet + BBRF	208.1	80.7	0.915	0.025	0.913	0.018	0.869	0.050
	HENet + BBRF	106.9	65.6	0.930	0.031	0.933	0.017	0.901	0.047
	LESOD + BBRF	99.1	57.0	0.926	0.029	0.912	0.021	0.897	0.049
OursII	CAVER + EDN	137.3	83.2	0.922	0.024	0.903	0.019	0.873	0.049
	PICRNet + EDN	155.5	78.6	0.916	0.026	0.904	0.018	0.870	0.050
	HENet + EDN	54.3	63.5	0.932	0.025	0.935	0.018	0.904	0.046
	LESOD + EDN	46.5	54.9	0.924	0.029	0.916	0.019	0.898	0.050
	CAVER + BBRF	189.9	85.3	0.927	0.022	0.913	0.017	0.875	0.046
	PICRNet + BBRF	208.1	80.7	0.920	0.024	0.917	0.016	0.873	0.048
	HENet + BBRF	106.9	65.6	0.935	0.023	0.936	0.017	0.901	0.046
	LESOD + BBRF	99.1	57.0	0.923	0.030	0.924	0.018	0.899	0.047

4.3.2 Qualitative Comparison. The visual comparisons in Fig. 7 illustrate our method against several representative SOTA RGB-D SOD approaches under challenging scenarios. Overall, our method achieves more accurate and stable saliency predictions. In low-contrast (1st row), cluttered-background (2nd row), multi-object (3rd row), and varying-illumination scenes (4th row), our method produces cleaner and more complete saliency maps by effectively leveraging reliable depth cues. Moreover, when depth information is limited or unreliable (5th and 6th rows), the proposed contribution-aware mechanism suppresses misleading depth interference and improves object delineation. For small-object cases (7th row), our method better preserves fine-grained structures and object integrity, further demonstrating its robustness. Additionally, Fig. 8 reports the consistency of saliency prediction across multiple models, demonstrating that the proposed method leads to more stable and reliable channel contribution assessment.

4.4 Ablation Analysis

4.4.1 Different Saliency Detection Tools. Since the final saliency maps produced by the contribution-aware fusion module (Sec. 3.4) depend on the employed RGB and RGB-D saliency detection tools, we further investigate the robustness of our framework under different tool selections. Specifically, we evaluate several representative RGB and RGB-D saliency detection models used in our method. EDN [52] and BBRF [31] are adopted as RGB saliency detection tools, while CAVER [34], PICRNet [12], as well as more recent and lightweight RGB-D models such as HENet [17] and LESOD [68], are employed as RGB-D saliency detection tools.

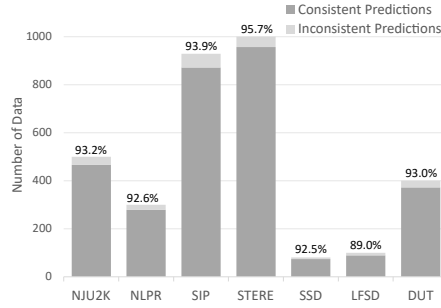


Fig. 8. Saliency prediction classification results for the same category samples (high/low-contribution) and misclassified samples on the GateNet [65], EDN [52] and BBRF [31]. “Consistent predictions” refer to cases where feeding an RGB-D pair into three models results in the same contribution assessment for RGB or depth channels based on saliency changes.

Table 4. Ablation Analysis of Threshold T_C in Eq. 4.

Tc Threshold	NJU2K		NLPR		LFSD	
	$F\omega_\beta \uparrow$	MAE $\downarrow$	$F\omega_\beta \uparrow$	MAE $\downarrow$	$F\omega_\beta \uparrow$	MAE $\downarrow$
0.2	0.916	0.027	0.909	0.019	0.860	0.054
0.3	0.918	0.026	0.910	0.019	0.863	0.053
0.4	0.915	0.027	0.908	0.020	0.862	0.054
Our	0.920	0.024	0.912	0.018	0.865	0.051

Table 5. Ablation analysis of different combinations of depth channel with RGB channels.

Dataset	NJU2K		NLPR		LFSD	
Metric	$F\omega_\beta \uparrow$	MAE $\downarrow$	$F\omega_\beta \uparrow$	MAE $\downarrow$	$F\omega_\beta \uparrow$	MAE $\downarrow$
One-Channel	0.919	0.026	0.909	0.022	0.862	0.053
Two-Channel	0.920	0.024	0.912	0.018	0.865	0.051
Three-Channel	0.910	0.026	0.902	0.024	0.861	0.052

The results in Table 3 show that, under both coarse-grained (“OursI”) and fine-grained (“OursII”) channel contribution assessment, different combinations of RGB and RGB-D tools lead to comparable performance trends and consistently outperform their corresponding baseline models, including CAVER, PICRNet, HENet, and LESOD (RGB-D tools), as well as EDN and BBRF (RGB tools evaluated using only RGB inputs). These results indicate that the proposed framework is robust to the choice of saliency detection tools and can achieve consistent performance improvements without relying on a specific backbone or a particular baseline model.

4.4.2 Ablation Analysis of Threshold T_C in Eq. 4. T_C calculates the noise disruption level corresponding to threshold exceedance. Our dynamic threshold is strategically set at half the range between the maximum and minimum values of the S-measure, typically falling within the range of 0.25 to 0.3, based on empirical findings. By statistically assessing the significance change induced by noise disruptions, we establish several fixed thresholds for comparison with our dynamic thresholds in Table 4. Results demonstrate the superior performance of dynamic thresholds over predetermined ones. This superiority arises from the dynamic threshold’s adaptive adjustment to noise disruption levels, leading to more accurate results.

4.4.3 Different Saliency Sensing Tool G . We evaluated the robustness of the saliency sensing tool (Eq. 3) under progressively increasing perturbations by testing three notable RGB saliency

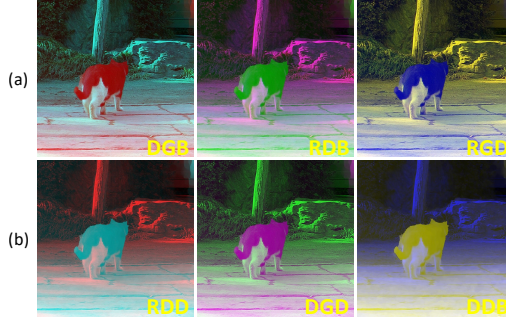


Fig. 9. Visualization of RGB channels replaced by one D channel (a) and two D channels (b).

Table 6. Ablation studies of different types of noise disruptions.

Dataset	NJU2K		NLPR		LFSD	
Metric	$F\omega_{\beta}\uparrow$	MAE $\downarrow$	$F\omega_{\beta}\uparrow$	MAE $\downarrow$	$F\omega_{\beta}\uparrow$	MAE $\downarrow$
Gaussian	0.901	0.033	0.898	0.023	0.852	0.056
Lighting	0.893	0.041	0.884	0.030	0.836	0.068
Compress	0.897	0.032	0.890	0.027	0.847	0.089
Salt-noisy	0.920	0.024	0.912	0.018	0.865	0.051

models: GateNet [65], EDN [52], and BBRF [31]. Our methodology involved processing RGB-D pairs from diverse datasets with these models to examine the consistency of channel contribution assessment under saliency variations. The experiments were conducted on seven benchmark datasets: NJU2K, NLPR, SIP, STERE, SSD, LFSD, and DUT. The results revealed high consistency in channel contribution assessment across these datasets, with consistency rates of 93.2%, 92.6%, 93.9%, 95.7%, 92.5%, 89%, and 93%, respectively. These findings demonstrate the effectiveness of our approach in estimating channel contributions and its robustness under controlled perturbations across different saliency models.

4.4.4 Different Combinations of Depth Channel. To examine how different channel combinations respond to perturbations and how their contributions evolve accordingly (see Sec. 3.3), we employ a channel-wise early fusion strategy (Sec. 3.2) by replacing RGB channels with depth channels. In the ablation study, we consider one-channel, two-channel, and three-channel depth settings. As illustrated in Fig. 9-(a), replacing one RGB channel with depth yields DGB, RDB, and RGD visualization samples. In Fig. 9-(b), replacing two RGB channels with depth yields RDD, DGD, and DDR visualization samples. In addition, the three-channel setting corresponds to directly replacing all RGB channels with depth, i.e., DDD. The results in Table 5 show that the two-channel setting achieves the best performance among the three configurations.

This observation highlights the importance of preserving an informative comparative relation between RGB and depth cues when assessing channel contributions under perturbations. Using only one depth channel introduces limited depth information, whereas directly replacing all three channels with depth not only weakens the complementary role of RGB cues but also leads to worse performance than the one-channel setting. By contrast, the two-channel setting, i.e., single-channel RGB with two-channel depth, provides a more balanced configuration for contribution evaluation and perturbation-aware fusion.

4.4.5 Different Types of Noise Disruptions on Degrading Channel Contributions. As visualized in Fig. 10, we introduced various noise disruptions to the RGB and D channels, including salt-and-pepper noise, Gaussian blur, lighting variations, and compression. Salt-and-pepper noise was added



Fig. 10. Impact visualization of various types of noise disruptions on degrading channel contributions.

Table 7. Time, Params and FLOPs comparisons of key components used in our method. “OursI” is the coarse-grained channel contribution assessment (Sec. 3.3.1), and “OursII” is the fine-grained channel contribution assessment (Sec. 3.3.2).

Components		Time(s)	Params(M)	FLOPs(G)
Channel-wise Early Fusion (Sec. 3.2) ①		0.009	--	--
Contribution Assessment	Coarse-grained (Sec. 3.3.1) ②	0.442	21.8	25.84
	Fine-grained (Sec. 3.3.2) ③	0.236		
Contribution-aware Fusion (Sec. 3.4)	Existing SOTA RGB-D Tool, e.g., CAVER ④	0.148	93.7	31.61
	Existing SOTA RGB Tool, e.g., BBRF ⑤	0.061	74.4	27.86
	Contribution-wise Fusion ⑥	0.012	--	--
OursI	① + max{②, ④, ⑤} + ⑥	0.463	189.9	85.31
OursII	① + max{②, ④, ⑤} + ③ + ⑥	0.699		

based on the signal-to-noise ratio, Gaussian blur was implemented by adjusting the kernel standard deviation, lighting variations were simulated by adjusting brightness, and compression quality was varied to induce distortions. Table 6 shows that salt-and-pepper noise yields the most favorable fused results. Since this type of noise disruptions, when appropriately controlled based on the signal-to-noise ratio, can effectively disrupt the channel contributions in a balanced manner without significantly degrading the overall structural information of the image.

4.5 Limitations

1) While the contribution-aware fusion scheme (Sec. 3.4) leverages the saliency perturbation-based channel contribution assessment (Sec. 3.3) to enhance saliency detection by prioritizing channels with higher contribution scores, these scores are inherently relative. When both RGB and depth (D) channels are of low quality, the method may still assign relatively high contribution scores, which can result in suboptimal saliency performance due to the limitations of the input data.

2) The time composition of different components in our method is shown in Table 7, and Table 2 compares the FPS, parameter counts, and performance metrics with SOTA RGB-D saliency methods. For “OursI”, which adopts coarse-grained saliency perception, the inference time includes channel-wise early fusion, coarse-grained contribution assessment, and contribution-aware fusion. The overall runtime is determined by the most time-consuming stage among the RGB branch, RGB-D branch, and contribution assessment, resulting in a total time of 0.463s. Similarly, for “OursII”, fine-grained contribution assessment is introduced to enable more refined contribution modeling,

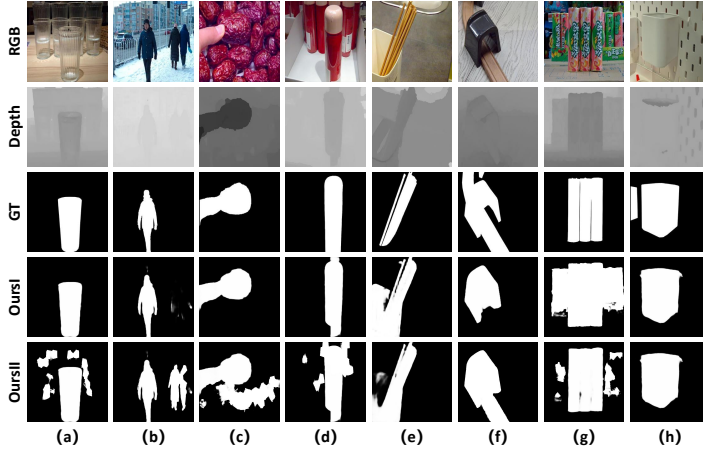


Fig. 11. Failure cases of the proposed method. From top to bottom, each column shows the input RGB image, the corresponding depth map, the ground-truth (GT), and the predictions of OursI and OursII.

leading to an inference time of 0.699s. The saliency perturbation-based channel contribution assessment (Sec. 3.3) accounts for the main computational overhead of the method, which may limit its applicability in real-time scenarios or on devices with limited computational resources.

3) As illustrated in Fig. 11, the proposed method may fail in certain challenging scenarios. In particular, when depth maps are severely degraded, ambiguous, or inconsistent with RGB appearance, the channel-wise contribution assessment may inaccurately estimate the reliability of depth information. This can result in incomplete object coverage or the introduction of false positive regions, especially in scenes with complex backgrounds, strong occlusions, thin structures, or multiple closely interacting objects. Although the fine-grained contribution assessment alleviates these issues to some extent, the method still relies on the quality and stability of depth cues. Addressing more robust depth reliability modeling and incorporating higher-level semantic or structural priors remain important directions for future work.

5 Conclusion

In conclusion, this paper has introduced a novel approach for salient object detection that addresses the challenges posed by complex scenes and varying quality of depth maps in RGB-D contexts. By focusing on fine-grained channel-wise contribution assessment, we have developed a method that accurately evaluates the contributions of different modality channels, such as RGB and depth (D), within diverse scene regions. This assessment guides a unique routing-style fusion scheme that optimizes the synergy between RGB and D channels without the need for additional labeling or learning. Our method achieves consistent and accurate salient object detection even in challenging scenarios, laying the groundwork for future research on more effective utilization of RGB and depth information. As for the future work, we are particularly interested in enhancing the computational efficiency of our method for real-time salient object detection. This includes optimizing speed and reducing resource consumption, making our approach suitable for time-sensitive and resource-constrained applications.

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